

ECON 21030: Honors Econometrics

Midterm 1

Spring Quarter 2023

There are a total of **120 points** on the exam. I recommend you spend about 20 minutes each on the first three problems and 30 minutes on the final problem. Use the remaining time to check your work and revisit parts you skipped.

1: [30 points] True, false, or it depends? If true, prove it or give an appropriate reference (briefly). If false, explain why or provide a counterexample. If it depends, explain why. No credit will be given without some justification, and good justification will be awarded full credit.

1. [5 points] If $Cov(X, Y) = Cov(Y, Z) = 0$, then $Cov(X, Z) = 0$.
2. [5 points] The law of iterated expectations implies that $E[Y] = E[E[Y|X]] = E[E[Y|log(X)]]$.
3. [5 points] For an iid sample (X_i, Y_i) of size N where $(X_i, Y_i) \sim N(\mu, \Sigma)$, $Var(\bar{X}_N - \bar{Y}_N) = \frac{1}{N}(\sigma_X^2 + \sigma_Y^2)$, where $\bar{X}_N$ is the sample mean of X_i and σ_X^2 is the variance of X_i (and analogously for Y_i).
4. [5 points] A level- α t -test of H_0 will reject with probability exactly equal to α when H_0 is true.
5. [5 points] An unbiased estimator of a given parameter is always the best choice.
6. [5 points] For a size N iid sample from a distribution with mean μ and variance $\sigma^2 < \infty$, then $E[\frac{1}{N} \sum_{i=1}^N X_i^2] = \mu^2$.

2: [25 points] Consider an iid sample $\{X_i\}_{i=1}^N$ from a uniform distribution over $[0, 1]$.

1. [5 points] Find a constant a such that $\bar{X}_N^2 = \frac{1}{N} \sum_{i=1}^N X_i^2 \xrightarrow{p} a$ and show that convergence in probability holds.
2. [5 points] Find the limiting distribution of $\sqrt{N}(\bar{X}_N^2 - a)$.
3. [5 points] Find the probability limit of $log(\bar{X}_N^2)$ (you do not need to evaluate the log).
4. [5 points] Find the limiting distribution of $\sqrt{N}(log(\bar{X}_N^2) - log(a))$.
5. [5 points] What is the length of a 95% confidence interval based on this approximation as a function of N ?

3: [25 points] Consider a model where individuals are endowed with productivity P_i . They can choose to acquire schooling, which costs C_i . Their wages are given by $Y_i = P_i + \beta S_i$, where S_i is a binary indicator for whether they acquired schooling. Assume (C_i, P_i) are distributed iid in the population.

1. [5 points] Assume individuals observe their C_i and P_i and know β . They chose to acquire schooling if the benefits in terms of wages outweigh the costs. Write down S_i as a function of C_i and (potentially) P_i .
2. [5 points] Write down expressions for average wages conditional on $S_i = 1$ and $S_i = 0$ in terms of conditional moments of P_i .
3. [5 points] Under what conditions will the difference in conditional means identify β ?
4. [10 points] Suppose that P_i and C_i are jointly normally distributed with marginal means μ_P and μ_C , correlation ρ and marginal variances of 1. Derive an expression for the mean of Y_i given $S_i = 1$ and $S_i = 0$ in terms of the marginal means of P_i and C_i , their correlation, and properties of the normal. How does the difference between β and the difference in conditional means depend on ρ ? (hint: you may need the fact that for a standard normal pdf, $-\phi'(x) = x\phi(x)$)

4: [40 points] Suppose you are interested in the following population regression:

$$Y_i = \alpha + \beta_1 X_i + \beta_2 W_i + \epsilon_i$$

However, you only observe the joint distribution of $(Y_i, X_i, \tilde{W}_i)$, where $\tilde{W}_i = W_i + u_i$, $u_i \sim N(0, \sigma_u^2)$ is an error-ridden version of W_i , with u_i independent of all other random variables. Hence you can only compute the coefficients in this alternative regression:

$$Y_i = \tilde{\alpha} + \tilde{\beta}_1 X_i + \tilde{\beta}_2 \tilde{W}_i + \tilde{\epsilon}_i$$

Your ultimate goal is to learn β_1 , which is core to your model. β_2 is not directly of interest.

1. [5 points] Derive an expression that relates β_1 to $\tilde{\beta}_1$ when $Cov(X_i, W_i) = 0$.
2. [5 points] Derive an expression that relates β_2 to $\tilde{\beta}_2$ when $Cov(X_i, W_i) = 0$. What is the role of σ_u^2 ?
3. [10 points] Now suppose that $Cov(X_i, W_i) = 1$, $Var(X_i) = 2$, and $Var(W_i) = 1$. Derive an explicit expression for β_1 in terms of the covariances of Y_i with X_i and W_i and the variance of X_i .
4. [10 points] Continuing to make the same assumptions, derive an expression for $\tilde{\beta}_1$ and explain why it differs from β_1 .
5. [5 points] Can you tell whether $\tilde{\beta}_1$ will be larger or smaller than β_1 in absolute value? How does this differ from the comparison between β_2 and $\tilde{\beta}_2$?
6. [5 points] What does that tell you about how measurement error affects the interpretation of regressors in a multivariate regression?